Raghd Boulos

Computer Engineering Student @ Technion

050-744-4557 | Raghd.boulos@gmail.com | github.com/RaghdB

GPA
89

EDUCATION

Technion – Israel Institute of Technology *Winter 2023/24 – Present*

B.Sc. in Computer Engineering, Faculty of Electrical Engineering

GPA: 89 · 6th semester

PROJECTS & RESEARCH

Research – Group Testing Algorithms — *with Prof. Nader H. Bshouty*

- Co-authored "Optimal Randomized Group Testing Algorithm to Determine the Number of Defectives".

Multi-threaded Web Server — *C, pthreads, Linux*

- Converted a single-threaded C web server into a concurrent one, using a **thread pool** and a **FIFO request queue**.
- Implemented **pthread synchronization**, writer-priority logging and UDP statistics queries.

Flappy Bird Game on FPGA — *SystemVerilog RTL, DE10-Lite*

- Designed and implemented a Flappy Bird-inspired game as **RTL in SystemVerilog**, featuring **VGA graphics**, animated sprites, day/night transitions and synchronized audio.
- Built physics-based movement, randomized obstacles, adaptive difficulty and scoring logic.

Digital Design Projects — *SystemVerilog RTL*

- Designed and verified RTL for a **64-bit ALU** and FSM-based control systems, including traffic-light controllers.

Heart-Rate Estimation from Video — *MATLAB, Signal Processing*

- Built a MATLAB **PPG signal-processing pipeline** to extract, filter and analyze a video signal for heart-rate estimation.

Course	Grade
Digital Systems and Computer Structure	89
Introduction to Systems Programming (MATAM)	88
Computer Organization And Programming (ATAM)	84
Data Structures 1	82
Introduction to Computer Science H	85
Theory of Electronic Circuits	91
Introduction to Probability H	96
Ordinary Differential Equations M	98
Physics 2Mm	100

SKILLS

Programming: C, C++, Python, JavaScript, SystemVerilog, MATLAB

Hardware: RTL design, FPGA, digital logic

Tools: Git, Bash, Linux, Windows

AI Tools: Claude / Codex, MCP, Skills, local LLMs

Languages: Arabic, Hebrew, English

WORK EXPERIENCE

Israel Railways — Push Notifications Publisher, Operational Control Center *Jan 2023 – Present*

- Publish real-time push notifications to passengers during live operations.
- Verify schedule accuracy across all railway information systems.

Initiatives beyond core responsibilities:

- Automated the push-message creation process using **Python**.
- Developed **JavaScript** tools that adapted legacy systems to modern standards.
- Wrote high-level design documents for system enhancements.

Ministry of Health — COVID-19 Call Center for Foreign Nationals (via Tikshoov) *Jan 2022 – Jan 2023*

- Guided uninsured callers and liaised with the Bikur Rofeh medical service; issued certificates.